Armoured Battle Console

Run the game on Windows or Linux with Docker

What this guide gives you: the full game — Village Battle (10 v 10 with computer players) and the Armoured Battle — running on your own computer with **one command**. Docker brings its own database, so you do not install or configure PostgreSQL, Python or anything else by hand.

You need	Details
A 64-bit computer	Windows 10 (22H2) or Windows 11, or any modern Linux (Ubuntu 22.04+, Debian 12, Fedora ...)
8 GB RAM, 5 GB free disk	The first start downloads about 1 GB (Python and PostgreSQL images)
Internet for the first start	After that the game runs offline on your network
A browser with WebGL	Chrome, Edge or Firefox (up to date). A real graphics card makes the 3D view smooth
The game folder	The `wargame` project folder (zip or git clone) with `Dockerfile` and `docker-compose.yml` inside

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1 Install Docker on Windows

Step 1. Turn on virtualisation

Open **Task Manager → Performance → CPU**. If it shows **Virtualization: Enabled**, go to step 2. If it says Disabled, restart into your BIOS/UEFI (usually F2, F10, Del or Esc during boot) and enable **Intel VT-x** / **AMD-V (SVM)**.

Step 2. Install WSL 2

Open **PowerShell as Administrator** (right-click Start → Terminal (Admin)) and run:

```
wsl --install
```

Restart the computer when it asks. If WSL is already installed, run `wsl --update` instead.

Step 3. Install Docker Desktop

Download **Docker Desktop for Windows** from `https://www.docker.com/products/docker-desktop/` and run the installer. Keep **"Use WSL 2 instead of Hyper-V"** ticked. Restart, then start Docker Desktop from the Start menu and wait until the whale icon in the taskbar says **Docker Desktop is running**.

> **Important:** Docker Desktop is free for personal use, education and small companies (fewer than 250 employees and less than $10 million revenue). Larger companies need a paid Docker subscription.

Step 4. Check it works

Open a new PowerShell window (normal, not admin) and run:

```
docker --version
docker compose version
```

Both commands should print a version number.

2 Install Docker on Linux

On Ubuntu / Debian / Fedora the official script installs Docker Engine and the Compose plugin:

```
curl -fsSL https://get.docker.com | sudo sh
sudo usermod -aG docker $USER      # run docker without sudo
newgrp docker                      # or log out and back in
```

Check it works:

```
docker --version
docker compose version
docker run --rm hello-world
```

> **Tip:** Use `docker compose` (with a space). The old `docker-compose` (with a dash) is version 1 and does not understand this project's file.

3 Start the game

Step 1. Get the game folder

Unzip the project or clone it. Put it somewhere simple, for example `C:\Games\wargame` on Windows or `~/wargame` on Linux.

Step 2. Open a terminal in that folder

Windows: open the folder in File Explorer, right-click an empty area → **Open in Terminal**.
Linux: `cd ~/wargame`.

You should see `docker-compose.yml` when you list the files (`dir` on Windows, `ls` on Linux).

Step 3. Start everything

```
docker compose up -d --build
```

The first time this takes 3–10 minutes: Docker downloads Python and PostgreSQL and builds the game image. Later starts take a few seconds. Watch the start-up with:

```
docker compose logs -f web
```

When you see `[wargame] ready → open http://localhost:8000` the game is up. Press **Ctrl+C** to stop watching the log (the game keeps running).

Step 4. Open the game

Go to `http://localhost:8000` in Chrome, Edge or Firefox.

> **Tip:** There is no database to set up. PostgreSQL runs in its own container and keeps its data in a Docker volume (`wargame_pgdata`), so your matches and settings survive restarts and reboots.

4 Play

1. On the home page click **Log in to start a Village Battle** and sign in with user `admin`, password `admin123` (change it — see section 7).

2. Click **Play Village Battle**. A new village is built and you join BLUE. 10 RED computer players hunt you; every empty BLUE place is filled by a computer team-mate.

3. Press **READY** in the lobby. Loot guns from the houses, grab a bike or car, and fight. Last squad standing wins.

Key	Action	Key	Action
W A S D	Move	Mouse	Look / aim
Shift	Sprint	Left click	Shoot
C	Crouch	Right click (hold)	Aim down sights
1 – 3	Switch gun	V	First / third person
F	Get in / out of a vehicle	Enter	Team chat

The Armoured Battle (tanks on real border terrain) is in the **Command Panel** after you log in.

5 Play with friends on your network

Friends on the same Wi-Fi / LAN can join your game from their own browser — they do not need Docker.

Step 1. Find your computer's IP address

Windows: run `ipconfig` and look for **IPv4 Address** (for example `192.168.1.25`). **Linux:** run `hostname -I`.

Step 2. Share the address, Room ID and key

When you start a Village Battle a message shows the **Room ID** and the **BLUE key**. Your friends open `http://192.168.1.25:8000` (your IP), type the Room ID, the key and a callsign, and click **Enter battle**. Up to 10 humans per side; computer players take every empty place.

Step 3. Allow it through the firewall (if friends cannot connect)

Windows: when Windows asks whether Docker Desktop may use the network, allow **Private networks**. Or run in an admin PowerShell:

```
New-NetFirewallRule -DisplayName "Wargame 8000" -Direction Inbound `
    -Protocol TCP -LocalPort 8000 -Action Allow -Profile Private
```

Linux with ufw: `sudo ufw allow 8000/tcp`

> **Important:** Only share the game on networks you trust. It is meant for a home or office LAN, not the open internet.

6 Everyday commands

Run these in the game folder (the one with `docker-compose.yml`).

What you want	Command
Start the game	`docker compose up -d`
Stop the game (keeps all data)	`docker compose down`
Is it running?	`docker compose ps`
See the game log	`docker compose logs -f web`
Restart after a crash	`docker compose restart web`
Update after new code	`docker compose up -d --build`
Wipe everything and start fresh	`docker compose down -v` then `docker compose up -d --build`

> **Important:** `down -v` deletes the database volume: all matches, replays and the admin password are gone. Only use it when you really want a clean start.

The game starts by itself when Docker starts (`restart: unless-stopped`). On Windows, enable **Docker Desktop → Settings → General → Start Docker Desktop when you sign in** if you want that after every reboot.

7 Settings (optional)

Create a file called `.env` in the game folder (next to `docker-compose.yml`) with any of these lines, then run `docker compose up -d` again:

```
ADMIN_USER=admin
ADMIN_PASSWORD=MyStrongPassword
WARGAME_PORT=8000
```

Setting	What it does
`ADMIN_PASSWORD`	Password of the admin account. Used only when that account is first created; to change it later run `docker compose exec web python manage.py changepassword admin`
`WARGAME_PORT`	Port on your computer. Use e.g. 8080 if something else already uses 8000; then open `http://localhost:8080`
`OFFLINE_TERRAIN=1`	Use only the elevation tiles that come with the game (no internet needed for the armoured theatres)
`LLM_BASE_URL`	Optional self-hosted AI model (Ollama etc.) for the AI advisor. Inside Docker, reach Ollama on your computer at `http://host.docker.internal:11434`

The database address is fixed inside `docker-compose.yml` on purpose, so a `DATABASE_URL` line in `.env` from a non-Docker setup cannot break the Docker setup.

8 Troubleshooting

Problem	Fix
“docker: command not found” or **“cannot connect to the Docker daemon”**	Docker Desktop is not running (Windows) — start it and wait for the whale icon. On Linux: `sudo systemctl start docker`, and make sure you logged out/in after `usermod -aG docker`.
“WSL 2 installation is incomplete” / Docker Desktop will not start	Run `wsl --update` in an admin PowerShell, restart, and check virtualisation is enabled (section 1, step 1).
“port is already allocated” / “address already in use”	Something else uses port 8000. Set `WARGAME_PORT=8080` in `.env` (section 7) and run `docker compose up -d`.
The page does not open right after starting	The first start migrates and seeds the database (about a minute). Watch `docker compose logs -f web` until it says **ready**.
“could not reach the database” in the log	Run `docker compose ps`. If `db` is not healthy, see `docker compose logs db`. As a last resort wipe and restart (section 6).
“exec docker/entrypoint.sh: no such file” on Windows	The file got Windows line endings. The image already fixes this; rebuild with `docker compose build --no-cache web`.
The 3D view is black or very slow	Update the browser and graphics driver. In Chrome check `chrome://gpu` shows WebGL as Hardware accelerated. Close other heavy tabs.
Friends cannot join	Use your LAN IP, not localhost; allow port 8000 in the firewall (section 5).
Still stuck	Send the output of `docker compose ps` and `docker compose logs --tail 100 web` to whoever looks after the game.